Practice workbook · Angles and areas of triangles inside a solid

Finding an angle or area of a triangle drawn inside a solid (a face or a slanted cross-section) by first finding its sides and then using trig or the cosine rule.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-18

1. Question 6 · part b · 2 marks

2023 NOV · Paper 01

Worked-example retry · Attempt part b only.

6

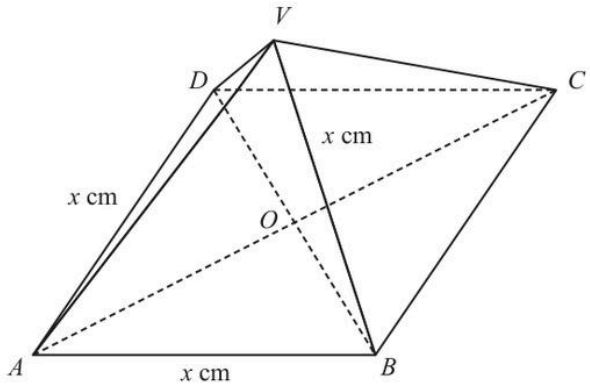


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with a horizontal square base.

$$AB = BC = CD = DA = x \text{ cm}$$

$$AV = BV = CV = DV = x \text{ cm}$$

O is the point of intersection of the diagonals of the base.
The vertex V of the pyramid is vertically above O

(a) Show that $VO = \frac{\sqrt{2}}{2}x$ cm

(3)

(b) Find, in degrees, the size of the angle AVC

(2)

(c) Find, in degrees to one decimal place, the size of the angle between the plane VAB and the plane VDC

(3)

The volume of the pyramid is 200 cm^3

Given that the volume of a pyramid $= \frac{1}{3} \times$ base area $\times$ height

(d) Find to 3 signficant figures, the value of x

(3)

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P 7 3 5 8 4 A 0 1 6 3 6

1. Question 6 · part b · 2 marks · Working and answer

Working and answer

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2. Question 8 · part d · 6 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

8

Diagram **NOT** accurately drawn

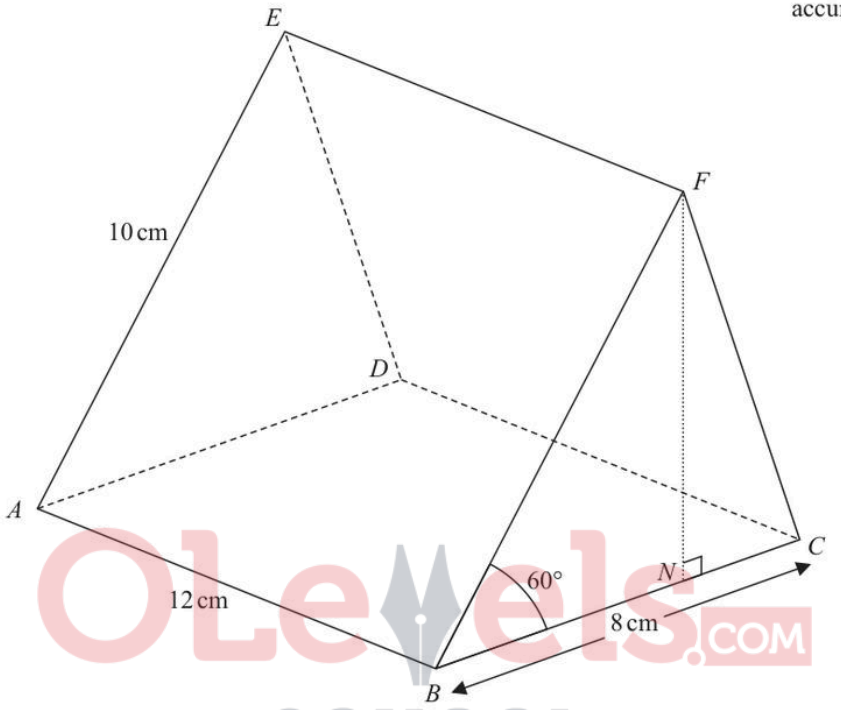


Figure 3

Figure 3 shows a right prism $ABCDEF$. The cross section BCF of the prism is a triangle.

$AB = DC = 12\,\text{cm}$ $BC = AD = 8\,\text{cm}$ $BF = AE = 10\,\text{cm}$ $\angle FBC = \angle EAD = 60°$

The point N lies on BC such that FN is perpendicular to BC.

(a) Show that $BN = 5\,\text{cm}$.

(2)

(b) Find, in cm to 3 significant figures, the length of EN.

(3)

The midpoint of BF is X and the midpoint of FC is Y.

(c) Find, in degrees to one decimal place, the size of the angle between the plane $ABCD$ and the plane $AXYD$.

(2)

(d) Find, in degrees to one decimal place, the size of the angle AYE.

(6)

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P 6 1 8 8 3 A 0 2 2 3 6

2. Question 8 · part d · 6 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 8 continued

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2. Question 8 · part d · 6 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 8 continued

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2. Question 8 · part d · 6 marks

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Further practice · Attempt part d only.

Question 8 continued

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(Total for Question 8 is 13 marks)

2. Question 8 · part d · 6 marks · Working and answer

Working and answer

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3. Question 4 · part b · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

4

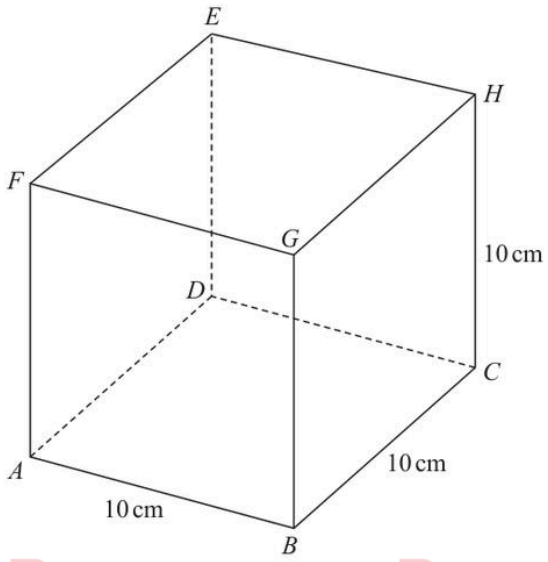


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Figure 2

Figure 2 shows a solid cube *ABCDEFGH* with sides of length 10 cm.

(a) Find, in cm, the exact length of *FC*

(2)

(b) Find, in degrees to one decimal place, the size of $\angle FCA$

(2)

(c) Find, to the nearest degree, the size of the obtuse angle between the plane *BFH* and the plane *BHC*

(5)

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3. Question 4 · part b · 2 marks · Working and answer

Working and answer

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4. Question 10 · part c · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

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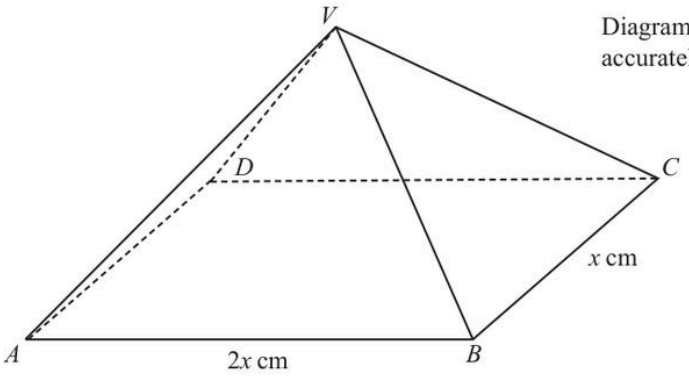


Figure 4

Figure 4 shows a right pyramid $ABCDV$

The base of the pyramid is a rectangle where,

$AB = DC = 2x$ cm $AD = BC = x$ cm

The edges VA, VB, VC and VD are all of equal length.

The angle between VA and $ABCD$ is 45°

(a) Show that $VA = \frac{\sqrt{10}}{2}x$ cm

(3)

(b) Find in cm, the exact height of the pyramid in terms of x

(2)

Find, in degrees to one decimal place,

(c) the size of angle VBA

(2)

(d) the size of the obtuse angle between the plane AVC and the plane BVD

(4)

Given that the volume of the pyramid is $9\sqrt{5}$ cm^3

(e) find the value of x

(2)

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4. Question 10 · part c · 2 marks · Working and answer

Working and answer

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5. Question 4 · part b · 3 marks

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Further practice · Attempt part b only.

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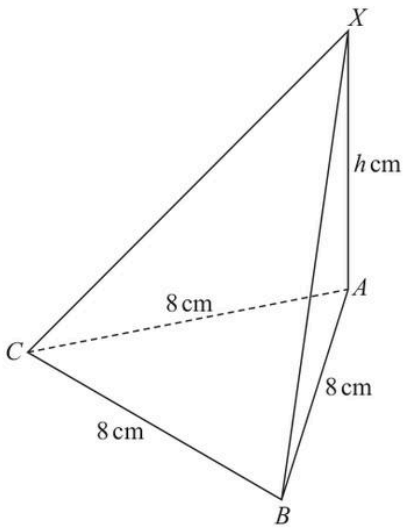


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a triangular pyramid $ABCX$.
The base ABC of the pyramid is an equilateral triangle where $AB = BC = CA = 8$ cm.
The vertex X of the pyramid is such that AX is perpendicular to the base of the pyramid and $AX = h$ cm.

The volume of the pyramid is $48\sqrt{3}$ cm^3

(a) Show that $h = 9$

(3)

(b) Find, in degrees to one decimal place, the size of angle BXC.

(3)

(c) Find, in degrees to one decimal place, the size of the angle between the plane BCX and the base ABC of the pyramid.

(3)

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5. Question 4 · part b · 3 marks · Working and answer

Working and answer

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6. Question 3 · part b · 3 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

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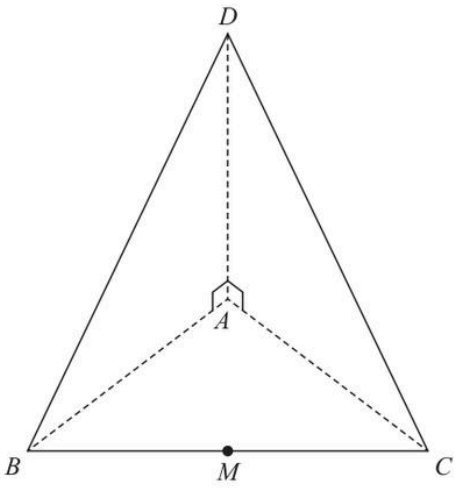


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangular pyramid $ABCD$.

The base, ABC, of the pyramid is a horizontal isosceles triangle with $AB = AC = 10\,\text{cm}$ and $BC = 16\,\text{cm}$. The midpoint of BC is M.

The face BCD of the pyramid is an isosceles triangle with $BD = CD = 26\,\text{cm}$ and D is vertically above A.

$\angle BAD = \angle CAD = 90°$

(a) Calculate the length, in cm, of AM.

(2)

Calculate, in degrees to the nearest degree,

(b) the size of $\angle BCD$,

(3)

(c) the size of the angle between the planes BCA and BCD.

(4)

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6. Question 3 · part b · 3 marks · Working and answer

Working and answer

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7. Question 8 · part c · 3 marks

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Further practice · Attempt part c only.

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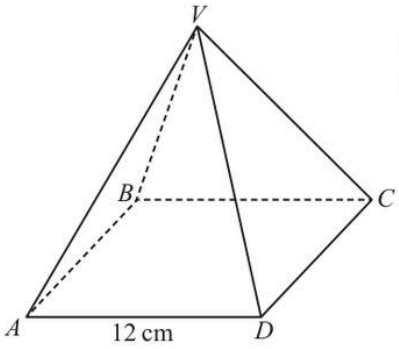


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with vertex V and square base, $ABCD$, of side 12 cm

The size of the angle between the plane VAB and the base $ABCD$ is $30°$

(a) Show that the height of the pyramid is $2\sqrt{3}$ cm

(2)

(b) Find, in cm, the exact length of VA

(3)

(c) Find, in cm^2, the exact area of triangle VAD

(3)

(d) Find, in cm, the exact length of the perpendicular from D to VA

(2)

(e) Find, in degrees to one decimal place, the size of the obtuse angle between the plane VAB and the plane VAD

(3)

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7. Question 8 · part c · 3 marks · Working and answer

Working and answer

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8. Question 9 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

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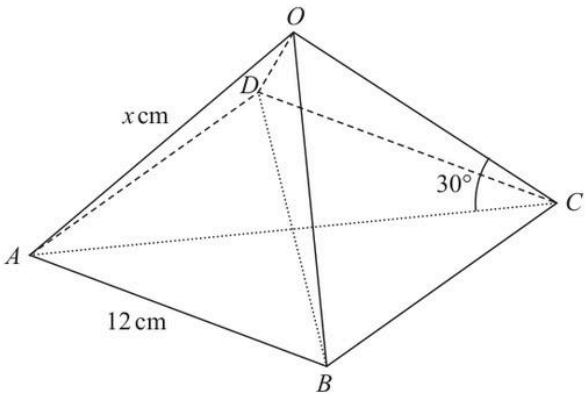


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the right pyramid $OABCD$ with a square base $ABCD$ of side 12 cm.

$OA = OB = OC = OD = x$ cm and $\angle OAC = \angle ODB = \angle OCA = \angle OBD = 30°$

(a) Find the exact length of AC

(2)

(b) Show that $x = 4\sqrt{6}$

(2)

(c) Find the total surface area, to the nearest cm^2, of the pyramid.

(5)

(d) Find the size of the obtuse angle, to the nearest degree, between the plane OAB and the plane OBC

(4)

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8. Question 9 · part c · 5 marks · Working and answer

Working and answer

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